

Decision Boundary & Score Dispersion
INTERACTIVE THRESHOLD

Drag the vertical discriminator line or use keyboard arrows to partition risk score bands.

Logistic Regression

Logistic (balanced)

Random Forest

XGBoost

XGBoost (balanced)

↑ CONFIRMED FRAUD DISTRIBUTION

↓ LEGITIMATE DISTRIBUTION

SCORE 0.00 (LEGITIMATE CONFIDENCE)

SCORE 0.50

SCORE 1.00 (FRAUDULENT ANOMALY)

Discriminator Threshold (τ):

0.14

Optimal F1

Catch 50% Fraud

Cap 5% Review Load

Default 0.50

69

TRUE POSITIVES (CAUGHT)

65.7% of genuine fraud identified

36

FALSE NEGATIVES (MISSED)

36 fraudulent leaks unprevented

346

FALSE POSITIVES (ALARMS)

8.6% false alert overhead

3,549

TRUE NEGATIVES (CLEAN)

91.1% of clean flow frictionless

16.6%

PRECISION (ALERT ACCURACY)

65.7%

RECALL (COVERAGE)

0.265

BALANCED F1 METRIC

104 /k

REVIEW LOAD PER 1,000 TXNS

Live Transaction Simulation & Feature Attribution

Instant forward-pass evaluation with localized linear Shapley-style attributions.

Load Sample Test Transaction

TRANSACTION AMOUNT (\$)

4850

FAILED TXNS (PAST 24H)

2

DISTANCE FROM USUAL LOCATION (KM)

720

MERCHANT CATEGORY

electronics

CHANNEL / TYPE

BANK_TRANSFER

DEVICE CATEGORY

desktop

PAYMENT INSTRUMENT

UPI

NEW DEVICE RECOGNIZED?

Yes (Unseen Hardware Fingerprint)

INTERNATIONAL ROUTE?

No (Domestic Route)

0.978

ASSIGNED RISK INDEX

CRITICAL RISK

Discriminator Verdict at $\tau = 0.14$: FLAG FOR IMMEDIATE MANUAL REVIEW

Top 5 Influential Risk Drivers (Feature Push Decomposition)



Financial Loss Curve & Operational Cost Minimizer DOLLAR OPTIMAL

Simulate total financial fallout based on manual review cost vs unrecovered fraudulent transaction losses.

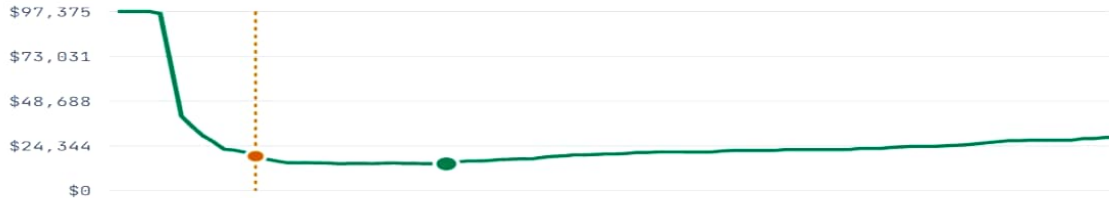
Manual Review Cost (per FP alert):

\$ 25

Avg Chargeback / Loss (per missed FN):

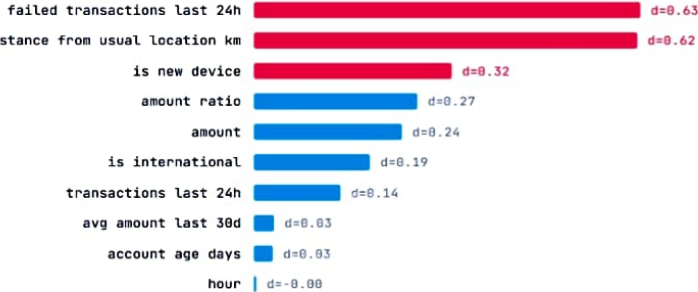
\$ 280

Dollar-optimal threshold is $\tau = 0.32$ (Minimum loss: **\$14,575**).
Current loss is **\$18,730** (+\$4,155 above optimal).



Cohen's d Separation Power

Standardized effect size separating genuine from fraudulent events.



Top Separation Signals: Density Profiles

Distance from typical geolocation and transaction values.

• Legitimate • Fraudulent

Distance from Home Geohash (km)



Transaction Amount Breakdown



Model Architecture Benchmark

Ranked by PR-AUC under strict 3% class prevalence.

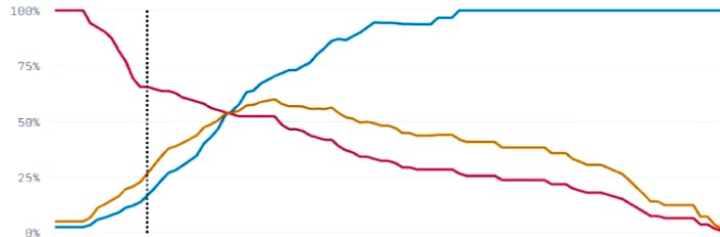
MODEL NAME	PR-AUC	ROC-AUC	BEST F1	OPT T	ACCURACY
Logistic Regression	0.1717	0.7502	0.601	0.32	97.0%
Logistic (balanced)	0.1700	0.7561	0.601	0.77	81.3%
Random Forest	0.1084	0.7290	0.573	0.37	96.9%
XGBoost	0.1030	0.7386	0.583	0.29	96.7%
XGBoost (balanced)	0.0724	0.6402	0.543	0.77	93.8%

Logistic Regression achieves the top PR-AUC (**0.1717**) over XGBoost (**0.1030**). Tree-based models overfit the sparse, 3% positive signal, while linear regularization maintains robust out-of-sample generalization.

Threshold Dynamics (Precision, Recall & F1)

Continuous curves across full [0.01 - 0.95] discriminant spectrum.

• Precision • Recall • F1 Score



Segment Prevalence Breakdown

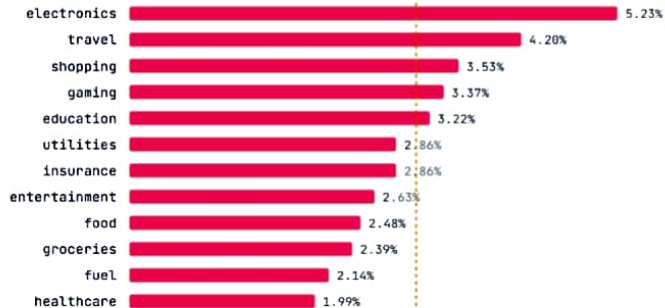
Relative risk divergence by merchant category and channel.

merchant category

transaction type

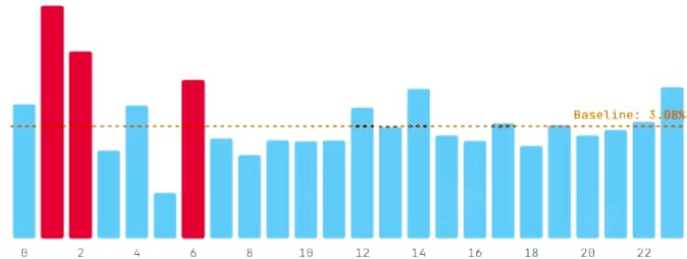
device type

payment method



Temporal Velocity: 24h Hourly Risk Index

Local clock hour showing nocturnal fraud spike patterns.





Held-out Evaluation Queue & Forensic Audit

Filterable transaction list evaluated at current threshold.

All Txns Caught (TP) Missed (FN) False Alarms (FP)

Search by ID, city, category, type...

Export Filtered Queue (.CSV)

STATUS	SCORE	ID	AMOUNT	TYPE	CATEGORY	LOCATION	DISTANCE	FAILED (24H)	DEVICE	INTL
CLEAN	0.064	TXN1000000	\$2,100	BANK_TRANSFER	fuel	Jaipur, RJ	10.5 km	0	Known	Dom
FALSE ALARM	0.185	TXN1000001	\$2,042	POS	fuel	Hyderabad, TS	1.2 km	1	Known	Dom
CLEAN	0.112	TXN1000002	\$1,486	BANK_TRANSFER	insurance	Bengaluru, KA	17.2 km	0	Known	Dom
FALSE ALARM	0.145	TXN1000003	\$1,353	POS	travel	Visakhapatnam, AP	18 km	1	Known	Dom
CLEAN	0.067	TXN1000004	\$5,116	ONLINE	healthcare	Delhi, DL	9 km	0	Known	Dom
CLEAN	0.058	TXN1000005	\$1,371	BANK_TRANSFER	travel	Nagpur, MH	14 km	0	Known	Dom
MISSED FRAUD	0.094	TXN1000006	\$7,603	ONLINE	shopping	Ahmedabad, GJ	5.2 km	0	Known	Dom
CLEAN	0.054	TXN1000007	\$2,976	POS	electronics	Kochi, KL	8.7 km	0	Known	Dom
CLEAN	0.053	TXN1000008	\$604	BANK_TRANSFER	utilities	Hyderabad, TS	2.5 km	0	Known	Dom
CLEAN	0.060	TXN1000009	\$2,368	BANK_TRANSFER	fuel	Chandigarh, CH	4.2 km	0	Known	Dom
CLEAN	0.050	TXN1000010	\$3,819	ONLINE	insurance	Guwahati, AS	7 km	0	Known	Dom
CLEAN	0.060	TXN1000011	\$403	BANK_TRANSFER	shopping	Lucknow, UP	5.5 km	0	Known	Dom